```
import numpy as np
import pandas as pd

import matplotlib.pyplot as plt
import seaborn as sns

from sklearn.model_selection import train_test_split
from sklearn.preprocessing import LabelEncoder
from sklearn.metrics import classification_report, roc_auc_score

from sklearn.ensemble import RandomForestClassifier
```

```
train_trans = pd.read_csv("train_transaction.csv")
train_id = pd.read_csv("train_identity.csv")
```

```
data = train_trans.merge(train_id, on="TransactionID", how="left")
```

```
data.shape
```

```
(3863, 434)
```

```
data.head()
```

	TransactionID	isFraud	TransactionDT	TransactionAmt	ProductCD	card1	c
0	2987000	0	86400	68.5	W	13926	
1	2987001	0	86401	29.0	W	2755	4
2	2987002	0	86469	59.0	W	4663	4
3	2987003	0	86499	50.0	W	18132	5
4	2987004	0	86506	50.0	H	4497	5

5 rows × 434 columns

```
data.info()
```

```
<class 'pandas.core.frame.DataFrame'>
```

```
RangeIndex: 3863 entries, 0 to 3862
Columns: 434 entries, TransactionID to DeviceInfo
dtypes: float64(399), int64(4), object(31)
memory usage: 12.8+ MB
```

```
missing_percentage = data.isnull().mean().sort_values(ascending=False)
missing_percentage.head(10)
```

	0
id_24	0.994305
id_22	0.994046
id_07	0.994046
id_21	0.994046
id_25	0.994046
id_26	0.994046
id_27	0.994046
id_23	0.994046
id_08	0.994046
D7	0.972560

dtype: float64

```
for col in data.columns:
    if data[col].dtype == 'object':
        data[col].fillna("Unknown", inplace=True)
    else:
        data[col].fillna(data[col].median(), inplace=True)
```

```
/tmp/ipykernel_161/3871526236.py:5: FutureWarning: A value is trying to be se
The behavior will change in pandas 3.0. This inplace method will never work b

For example, when doing 'df[col].method(value, inplace=True)', try using 'df.


  data[col].fillna(data[col].median(), inplace=True)
/tmp/ipykernel_161/3871526236.py:3: FutureWarning: A value is trying to be se
The behavior will change in pandas 3.0. This inplace method will never work b

For example, when doing 'df[col].method(value, inplace=True)', try using 'df.


  data[col].fillna("Unknown", inplace=True)
```

```
le = LabelEncoder()

for col in data.select_dtypes(include='object').columns:
    data[col] = le.fit_transform(data[col])
```

```
X = data.drop("isFraud", axis=1)
y = data["isFraud"]
```

```
X_train, X_test, y_train, y_test = train_test_split(
    X, y,
    test_size=0.2,
    stratify=y,
    random_state=42
)
```

```
sns.countplot(x=y)
plt.title("Fraud vs Non-Fraud Distribution")
plt.show()
```

```
sns.boxplot(x='isFraud', y='TransactionAmt', data=data)
plt.title("Transaction Amount vs Fraud")
plt.show()
```

```
from imblearn.over_sampling import SMOTE

smote = SMOTE(random_state=42)
X_train_res, y_train_res = smote.fit_resample(X_train, y_train)
```

```
print("Before SMOTE:")
print(y_train.value_counts())

print("\nAfter SMOTE:")
print(pd.Series(y_train_res).value_counts())
```

```
Before SMOTE:
isFraud
0    3032
1      58
Name: count, dtype: int64
```

```
After SMOTE:
isFraud
0    3032
1    3032
Name: count, dtype: int64
```

```
rf = RandomForestClassifier(
    n_estimators=200,
    max_depth=10,
    class_weight='balanced',
    random_state=42
)

rf.fit(X_train_res, y_train_res)
```

```
RandomForestClassifier
RandomForestClassifier(class_weight='balanced', max_depth=10, n_estimators=20
                       random_state=42)
```

```
rf_preds = rf.predict(X_test)
rf_probs = rf.predict_proba(X_test)[:, 1]

print(classification_report(y_test, rf_preds))
print("ROC-AUC Score:", roc_auc_score(y_test, rf_probs))
```

```
              precision    recall  f1-score   support

           0       0.99      1.00      0.99       758
           1       0.67      0.27      0.38        15

    accuracy                           0.98       773
   macro avg       0.83      0.63      0.69       773
weighted avg       0.98      0.98      0.98       773

ROC-AUC Score: 0.8569041336851363
```

```
feature_importance = pd.Series(
    rf.feature_importances_,
    index=X.columns
).sort_values(ascending=False)

feature_importance.head(10)
```

	0
V294	0.028283
card6	0.025453
V295	0.022332

V280	0.021723
C5	0.020951
C6	0.020134
V317	0.019671
V308	0.018842
V318	0.018623
C2	0.018412

dtype: float64

```
!pip install xgboost
from xgboost import XGBClassifier
```

```
Requirement already satisfied: xgboost in /usr/local/lib/python3.12/dist-pack
Requirement already satisfied: numpy in /usr/local/lib/python3.12/dist-packag
Requirement already satisfied: nvidia-nccl-cu12 in /usr/local/lib/python3.12/
Requirement already satisfied: scipy in /usr/local/lib/python3.12/dist-packag
```

```
xgb = XGBClassifier(
    n_estimators=300,
    max_depth=6,
    learning_rate=0.05,
    subsample=0.8,
    colsample_bytree=0.8,
    scale_pos_weight=10,
    random_state=42,
    eval_metric='logloss'
)

xgb.fit(X_train_res, y_train_res)
```

```
▾ XGBClassifier
XGBClassifier(base_score=None, booster=None, callbacks=None,
              colsample_bylevel=None, colsample_bynode=None,
              colsample_bytree=0.8, device=None, early_stopping_rounds=None,
              enable_categorical=False, eval_metric='logloss',
              feature_types=None, feature_weights=None, gamma=None,
              grow_policy=None, importance_type=None,
              interaction_constraints=None, learning_rate=0.05, max_bin=None,
              max_cat_threshold=None, max_cat_to_onehot=None,
              max_delta_step=None, max_depth=6, max_leaves=None,
              min_child_weight=None, missing=nan, monotone_constraints=None,
              multi_strategy=None, n_estimators=300, n_jobs=None,
```

```
xgb_preds = xgb.predict(X_test)
xgb_probs = xgb.predict_proba(X_test)[:, 1]

print(classification_report(y_test, xgb_preds))
print("ROC-AUC Score:", roc_auc_score(y_test, xgb_probs))
```

```
              precision    recall  f1-score   support

           0       0.99      1.00      0.99       758
           1       0.67      0.40      0.50        15

    accuracy                           0.98       773
   macro avg       0.83      0.70      0.75       773
weighted avg       0.98      0.98      0.98       773

ROC-AUC Score: 0.8186455584872472
```

```
results = pd.DataFrame({
    "Model": ["Random Forest", "XGBoost"],
    "ROC-AUC": [
        roc_auc_score(y_test, rf_probs),
        roc_auc_score(y_test, xgb_probs)
    ]
})

results
```

	Model	ROC-AUC
0	Random Forest	0.856904
1	XGBoost	0.818646

Next steps: Generate code with `results` New interactive sheet

```
xgb_importance = pd.Series(
    xgb.feature_importances_,
    index=X.columns
).sort_values(ascending=False)

xgb_importance.head(10)
```

	0
V70	0.126917
V30	0.098886
V29	0.047056
V309	0.037784

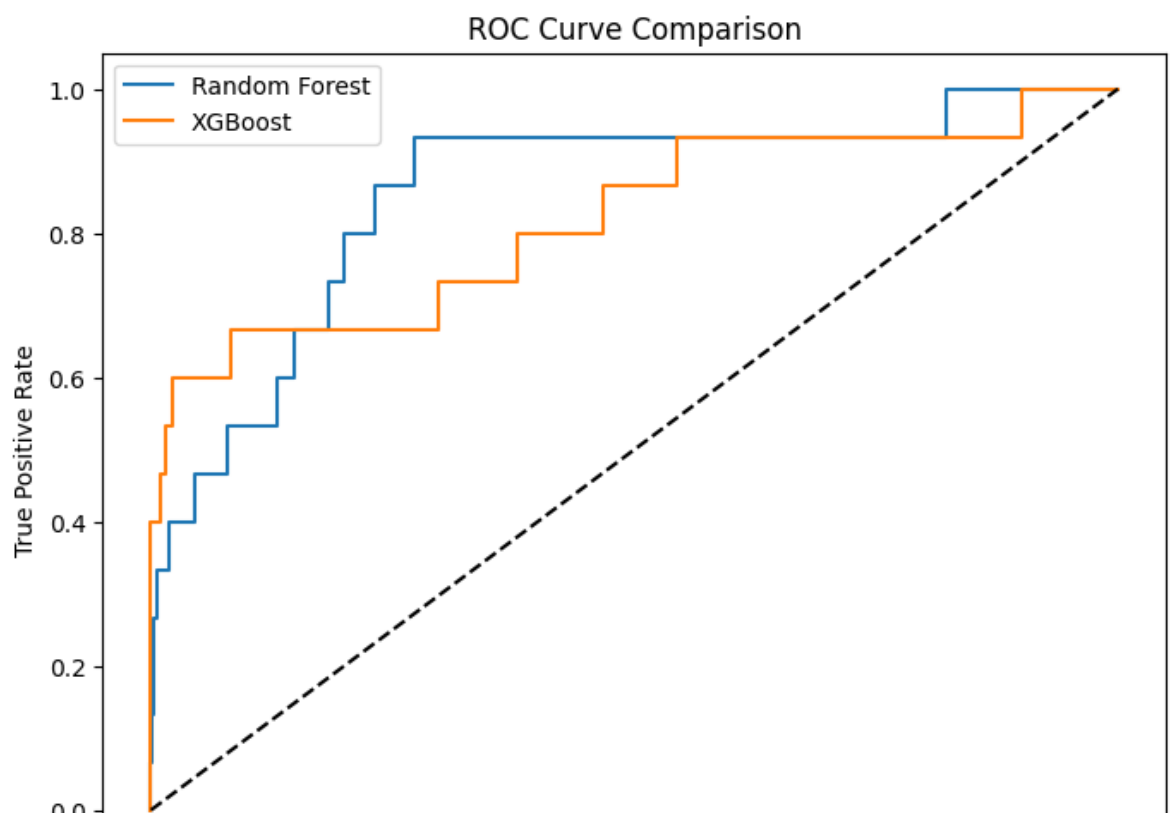
V160	0.037671
V274	0.029771
V292	0.026952
V126	0.019648
V155	0.016195
V287	0.016153

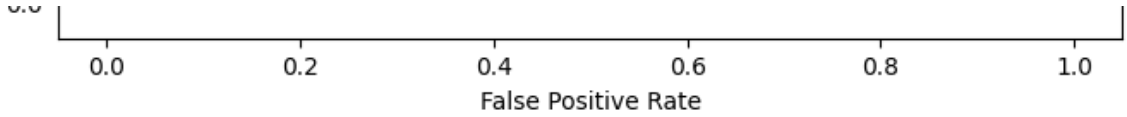
dtype: float32

```
from sklearn.metrics import roc_curve

fpr_rf, tpr_rf, _ = roc_curve(y_test, rf_probs)
fpr_xgb, tpr_xgb, _ = roc_curve(y_test, xgb_probs)

plt.figure(figsize=(8,6))
plt.plot(fpr_rf, tpr_rf, label="Random Forest")
plt.plot(fpr_xgb, tpr_xgb, label="XGBoost")
plt.plot([0,1], [0,1], 'k--')
plt.xlabel("False Positive Rate")
plt.ylabel("True Positive Rate")
plt.title("ROC Curve Comparison")
plt.legend()
plt.show()
```





XGBoost outperforms Random Forest with a higher ROC-AUC and recall score. Since fraud detection prioritizes identifying fraudulent transactions even at the cost of some false positives, XGBoost is better suited for this problem.

✓ *False Positives*

Legitimate customers blocked

Poor user experience

✓ *False Negatives*

Fraud goes undetected

Financial and reputational loss

✓ *Ethical Challenge*

Bias in training data may unfairly flag specific groups. Continuous auditing and transparency are essential before deployment.

```
sample = pd.read_csv("sample_submission.csv")
sample.head()
```

	TransactionID	isFraud
0	3663549	0.5
1	3663550	0.5
2	3663551	0.5

3	3663552	0.5
4	3663553	0.5

```
from google.colab import files
files.download("sample_submission.csv")
```

```
!ls
```

```
ieee-fraud-detection.zip  sample_submission.csv  train_identity.csv
sample_data               test_transaction.csv   train_transaction.csv
```

```
import pandas as pd

# Load test files
test_trans = pd.read_csv("test_transaction.csv")
test_id = pd.read_csv("test_identity.csv")

# Merge
test_data = test_trans.merge(test_id, on="TransactionID", how="left")

# Handle missing values
for col in test_data.columns:
    if test_data[col].dtype == 'object':
        test_data[col] = test_data[col].fillna("Unknown")
    else:
        test_data[col] = test_data[col].fillna(test_data[col].median())

# Encode categorical columns
for col in test_data.select_dtypes(include='object').columns:
    test_data[col] = test_data[col].astype('category').cat.codes
```

```
xgb.feature_names_in_
```

```
array(['TransactionID', 'TransactionDT', 'TransactionAmt', 'ProductCD',
       'card1', 'card2', 'card3', 'card4', 'card5', 'card6', 'addr1',
       'addr2', 'dist1', 'dist2', 'P_emaildomain', 'R_emaildomain', 'C1',
       'C2', 'C3', 'C4', 'C5', 'C6', 'C7', 'C8', 'C9', 'C10', 'C11',
       'C12', 'C13', 'C14', 'D1', 'D2', 'D3', 'D4', 'D5', 'D6', 'D7',
       'D8', 'D9', 'D10', 'D11', 'D12', 'D13', 'D14', 'D15', 'M1', 'M2',
       'M3', 'M4', 'M5', 'M6', 'M7', 'M8', 'M9', 'V1', 'V2', 'V3', 'V4',
       'V5', 'V6', 'V7', 'V8', 'V9', 'V10', 'V11', 'V12', 'V13', 'V14',
       'V15', 'V16', 'V17', 'V18', 'V19', 'V20', 'V21', 'V22', 'V23',
       'V24', 'V25', 'V26', 'V27', 'V28', 'V29', 'V30', 'V31', 'V32',
       'V33', 'V34', 'V35', 'V36', 'V37', 'V38', 'V39', 'V40', 'V41',
       'V42', 'V43', 'V44', 'V45', 'V46', 'V47', 'V48', 'V49', 'V50',
       'V51', 'V52', 'V53', 'V54', 'V55', 'V56', 'V57', 'V58', 'V59',
```

```
       'V60', 'V61', 'V62', 'V63', 'V64', 'V65', 'V66', 'V67', 'V68',
       'V69', 'V70', 'V71', 'V72', 'V73', 'V74', 'V75', 'V76', 'V77',
       'V78', 'V79', 'V80', 'V81', 'V82', 'V83', 'V84', 'V85', 'V86',
       'V87', 'V88', 'V89', 'V90', 'V91', 'V92', 'V93', 'V94', 'V95',
       'V96', 'V97', 'V98', 'V99', 'V100', 'V101', 'V102', 'V103', 'V104',
       'V105', 'V106', 'V107', 'V108', 'V109', 'V110', 'V111', 'V112',
       'V113', 'V114', 'V115', 'V116', 'V117', 'V118', 'V119', 'V120',
       'V121', 'V122', 'V123', 'V124', 'V125', 'V126', 'V127', 'V128',
       'V129', 'V130', 'V131', 'V132', 'V133', 'V134', 'V135', 'V136',
       'V137', 'V138', 'V139', 'V140', 'V141', 'V142', 'V143', 'V144',
       'V145', 'V146', 'V147', 'V148', 'V149', 'V150', 'V151', 'V152',
       'V153', 'V154', 'V155', 'V156', 'V157', 'V158', 'V159', 'V160',
       'V161', 'V162', 'V163', 'V164', 'V165', 'V166', 'V167', 'V168',
       'V169', 'V170', 'V171', 'V172', 'V173', 'V174', 'V175', 'V176',
       'V177', 'V178', 'V179', 'V180', 'V181', 'V182', 'V183', 'V184',
       'V185', 'V186', 'V187', 'V188', 'V189', 'V190', 'V191', 'V192',
       'V193', 'V194', 'V195', 'V196', 'V197', 'V198', 'V199', 'V200',
       'V201', 'V202', 'V203', 'V204', 'V205', 'V206', 'V207', 'V208',
       'V209', 'V210', 'V211', 'V212', 'V213', 'V214', 'V215', 'V216',
       'V217', 'V218', 'V219', 'V220', 'V221', 'V222', 'V223', 'V224',
       'V225', 'V226', 'V227', 'V228', 'V229', 'V230', 'V231', 'V232',
       'V233', 'V234', 'V235', 'V236', 'V237', 'V238', 'V239', 'V240',
       'V241', 'V242', 'V243', 'V244', 'V245', 'V246', 'V247', 'V248',
       'V249', 'V250', 'V251', 'V252', 'V253', 'V254', 'V255', 'V256',
       'V257', 'V258', 'V259', 'V260', 'V261', 'V262', 'V263', 'V264',
       'V265', 'V266', 'V267', 'V268', 'V269', 'V270', 'V271', 'V272',
       'V273', 'V274', 'V275', 'V276', 'V277', 'V278', 'V279', 'V280',
       'V281', 'V282', 'V283', 'V284', 'V285', 'V286', 'V287', 'V288',
       'V289', 'V290', 'V291', 'V292', 'V293', 'V294', 'V295', 'V296',
       'V297', 'V298', 'V299', 'V300', 'V301', 'V302', 'V303', 'V304',
       'V305', 'V306', 'V307', 'V308', 'V309', 'V310', 'V311', 'V312',
       'V313', 'V314', 'V315', 'V316', 'V317', 'V318', 'V319', 'V320',
       'V321', 'V322', 'V323', 'V324', 'V325', 'V326', 'V327', 'V328',
       'V329', 'V330', 'V331', 'V332', 'V333', 'V334', 'V335', 'V336',
       'V337', 'V338', 'V339', 'id_01', 'id_02', 'id_03', 'id_04',
       'id_05', 'id_06', 'id_07', 'id_08', 'id_09', 'id_10', 'id_11',
       'id_12', 'id_13', 'id_14', 'id_15', 'id_16', 'id_17', 'id_18',
       'id_19', 'id_20', 'id_21', 'id_22', 'id_23', 'id_24', 'id_25',
       'id_26', 'id_27', 'id_28', 'id_29', 'id_30', 'id_31', 'id_32',
       'id_33', 'id_34', 'id_35', 'id_36', 'id_37', 'id_38', 'DeviceType',
       'DeviceInfo'], dtype='<U14')
```

```
# 1. Add missing columns
for col in xgb.feature_names_in_:
    if col not in test_data.columns:
        test_data[col] = 0

# 2. Drop extra columns
test_data = test_data[xgb.feature_names_in_]
```

```
/tmp/ipykernel_161/2097303977.py:4: PerformanceWarning: DataFrame is highly f
  test_data[col] = 0
/tmp/ipykernel_161/2097303977.py:4: PerformanceWarning: DataFrame is highly f
  test_data[col] = 0
```

```
/tmp/ipykernel_161/2097303977.py:4: PerformanceWarning: DataFrame is highly f
  test_data[col] = 0
/tmp/ipykernel_161/2097303977.py:4: PerformanceWarning: DataFrame is highly f
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/tmp/ipykernel_161/2097303977.py:4: PerformanceWarning: DataFrame is highly f
  test_data[col] = 0
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  test_data[col] = 0
/tmp/ipykernel_161/2097303977.py:4: PerformanceWarning: DataFrame is highly f
  test_data[col] = 0
/tmp/ipykernel_161/2097303977.py:4: PerformanceWarning: DataFrame is highly f
  test_data[col] = 0
```

```
test_preds = xgb.predict_proba(test_data)[:, 1]
```

```
submission = pd.DataFrame({
    "TransactionID": test_trans["TransactionID"],
    "isFraud": test_preds
})

submission.to_csv("submission.csv", index=False)
```

```
from google.colab import files
files.download("submission.csv")
```

Start coding or generate with AI.